Chemical Reactions and Equations

1. Chemical Equations

- **Writing a Chemical Equation**
  - Word Equations: Represent reactions using words.
    - Example: Magnesium + Oxygen → Magnesium Oxide
  - Chemical Equations: Use symbols and formulas.
    - Example: $Mg + O_2 \rightarrow MgO$
- **Balanced Chemical Equations**
  - Law of Conservation of Mass: Mass remains constant in a reaction
  - Balancing Steps:
    - 1. Draw Boxes: Around each formula.
    - 2. List Atoms: Count elements on both sides.
    - 3. Choose Element with Maximum Atoms: Start balancing from the most complex compound.
    - 4. Balance Oxygen first: Example: $Fe + H_2O \rightarrow Fe_2O_3 + H_2$; $Fe + 4H_2O \rightarrow Fe_2O_3 + 4H_2$
    - 5. Balance Hydrogen: Adjust coefficients to balance H atoms.
    - 6. Balance Iron: Adjust Fe atoms to balance the equation.
    - 7. Include Physical States: Indicate (s), (l), (g) and (aq).
    - 8. Final Balanced Equation: $3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$.

2. Types of Chemical Reactions

- **Combination Reaction**
  - Definition: Two/more reactants form one product.
  - Example: $CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + \text{Heat}$
- **Decomposition Reaction**
  - Definition: One compound breaks into simpler substances.
  - Example: $2FeSO_4(s) \xrightarrow{\Delta} Fe_2O_3(s) + SO_2(g) + SO_3(g)$
  - Types:
    1. Thermal Decomposition: Uses Heat
       Example: $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$
    2. Electrolytic Decomposition: Uses electricity
       Example: $2H_2O(l) \xrightarrow{\text{electric current}} 2H_{2(g)} + O_{2(g)}$
    3. Photodecomposition: Uses light
       Example: $2AgCl \xrightarrow{\text{light}} 2Ag + Cl_2$
- **Displacement Reaction**
  - Definition: A more reactive element displaces a less reactive one.
  - Example: $Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$
- **Double Displacement Reaction**
  - Definition: Exchange of ions between reactants.
  - Example: $Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$
  - Precipitation: Formation of an insoluble substance (e.g., $BaSO_4$).
- **Oxidation and Reduction**
  - Oxidation: Gain of oxygen/loss of hydrogen.
  - Reduction: Loss of oxygen/gain of hydrogen.
  - Examples:
    - $2Cu + O_2 \xrightarrow{\Delta} 2CuO$ (oxidation)
    - $CuO + H_2 \xrightarrow{\Delta} Cu + H_2O$ (reduction)
  - Redox Reaction: Oxiation and reduction occurring simultaneously
    - $ZnO + C \rightarrow Zn + CO$ (ZnO reduced, C oxidised)

3. Have you observed the effects of oxidation reactions in everyday life?

- **Corrosion**
  - Definition: Metal deterioration due to environmental factors.
  - Examples:
    - Rusting of iron (reddish-brown coat).
    - Black coat on silver green coat on copper.
  - Impact: Damages bridges, ships, and iron objects.
- **Rancidity**
  - Definition: Oxidation of fats/oils causing bad taste/smell.
  - Prevention:
    - Use antioxidants.
    - Store in airtight containers.
    - Flush chips bags with nitrogen gas.

4. Energy Changes in a Reaction

- **Exothermic**
  - Definition: Heat evolves in a reaction
  - Example: Respiration: $C_6H_{12}O_6(aq) + 6O_2(aq) \rightarrow 6CO_2(aq) + 6H_2O(l) + \text{Energy}$
- **Endothermic**
  - Definition: Heat is absorbed in a reaction
  - Example: $CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g)$

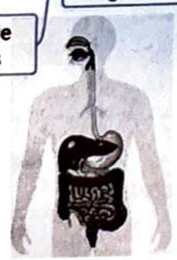
Life Processes

1. What Are Life Processes?

- **Maintenance in Living Organisms:** Prevents breakdown, active at rest.
- **Energy Requirement:** Needs external food sources.
- **Molecular Movements:** Essential for growth, powered by energy.
- **Specialised Systems:** Tissues manage nutrition, respiration, excretion. Transport systems distribute nutrients, remove waste.

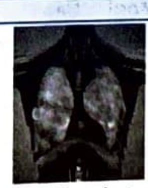
2. Nutrition

- **Autotrophic Nutrition**
  - **Energy and Material Needs:** Food fuels movement, growth, maintenance.
  - **Photosynthesis:** Light absorption → Energy conversion → CO_2 to carbohydrates.
  - **Stomata:** Gaseous exchange regulated by guard cells.
  - **Nutrient Uptake:** Water and soil, minerals (e.g., nitrogen for proteins).
- **Heterotrophic Nutrition**
  - Absorb decomposed matter (fungi), ingest food (humans), parasitic (tapeworms).
- **Nutrition in Organisms**
  - **Unicellular:** Absorption, photosynthesis, ingestion (Amoeba).
  - **Multicellular:** Specialised digestion (humans).
- **Nutrition in Human Beings**
  - Alimentary Canal
    - Tube (mouth to anus) for digestion and absorption.
  - Digestive Process
    - **Ingestion:** Chewing, salivary amylase.
    - **Stomach:** HCl (acidic), pepsin (proteins), mucus (lining protection).
    - **Small Intestine:** Bile (fat emulsification), pancreatic & intestinal enzymes (nutrient conversion).
    - **Absorption:** Villi absorb nutrients, large intestine absorbs water, excretes waste.



3. Respiration

- Cellular Respiration
  - **Aerobic:** $O_2 \rightarrow CO_2 + H_2O +$ energy
  - **Anaerobic:** No O_2 (e.g., lactic acid in muscles)
- ATP – The Energy Currency
  - Stores energy for cell activities.
- Gas Exchange
  - **Plants:** Stomata (CO_2 in, O_2 out).
  - **Animals:** Specialised respiratory organs.
- Human Respiratory System



  - **Pathway:** Nostrils → Throat → Lungs → Alveoli (gas exchange)
  - **Breathing:** Inhale (O_2 in), Exhale (CO_2 out)
  - **Oxygen Transport:** Haemoglobin carries O_2, CO_2 dissolves in blood

4. Transportation

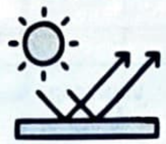
Human Circulatory System

- Transportation in Human Beings
  - Blood Composition: Plasma (nutrients/wastes), RBCs (O_2 transport).
  - The Heart: Muscular pump, four chambers.
  - Double Circulation: Blood passes heart twice per cycle.
  - Blood Pressure: Systolic: 120 mm Hg, Diastolic: 80 mm Hg.
  - Blood Vessels: Arteries except pulmonary artery (oxygenated, high pressure), Veins except pulmonary vein (deoxygenated, low pressure Capillaries (exchange of substances).
  - Platelets and Clotting: Prevents blood loss.
  - Lymph: Transports fats, fluids.
- Transportation in Plants
  - Water and Mineral Absorption: Active root uptake.
  - Water Transport (Xylem): Moves water via transpiration pull.
  - Food Transport (Phloem): Translocates food via osmotic pressure.

5. Excretion

- Excretion in Human Beings
  - Excretory System: **Organs:** Kidneys, ureters, bladder, urethra.
  - Urine Formation: Nephron filtration → Urine excretion.
  - Artificial Kidney (Dialysis): Filters blood in kidney failure.
- Excretion in Plants
  - O_2, H_2O via stomata; waste in vacuoles or shed with leaves.

Light - Reflection and Refraction

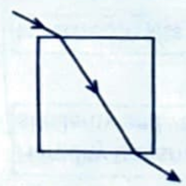


1. Reflection of Light
- **Definition**
  - Bouncing of light when it strikes a polished surface.
- **Laws of Reflection**
  - Angle of incidence = Angle of reflection.
  - Incident ray, reflected ray and normal lies in the same plane.

2. Spherical Mirrors
- **Types**
  - **Concave Mirror:** Reflecting surface curved inward.
  - **Convex Mirror:** Reflecting surface curved outward.
- **Image Formation by Spherical Mirrors**
  - Concave Mirror
    - Forms real/inverted or virtual/erect images depending on object position. Real image may be magnified or diminished. Virtual image is always erect and enlarged.
  - Convex Mirror
    - Always forms virtual, erect, and diminished images.
  - Ray Diagrams for Image Formation
    - **Parallel Ray Rule (Principal):** A ray parallel to the axis passes through (or appears to diverge from) Focus(F).
    - **Focus Ray Rule:** A ray passing through Focus (F) emerges parallel to principal axis
    - **Centre of Curvature Rule:** A ray passing through C reflects back on the same path.
    - **Pole Rule:** A ray striking the pole follows the law of reflection.
- **Uses of Mirrors**
  - Concave Mirrors
    - Used in torches, shaving mirrors, solar furnaces and dentists' mirrors.
  - Convex Mirrors
    - Used in vehicle rear-view mirrors for a wider field of view.
- **Mirror Formula and Magnification**
  - Mirror Formula $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$
  - Magnification Formula $m = \frac{h'}{h} = -\frac{v}{u}$
  - Magnification is **positive for virtual images, and negative for real images.**

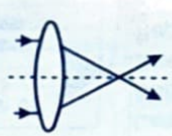
3. Sign Conventions for Mirrors (New Cartesian Sign Convention)
- **Distance Measurement**
  - **Parallel to Axis:** Measured from the pole (P); in the direction of incident light is positive, opposite to the direction of incident light is negative.
  - **Perpendicular to Axis:** Above principal axis is positive, below is negative.
- **Focal Length and Image Properties**
  - **Concave Mirror:** Focal length (f) and radius of curvature (R) are negative.
  - **Convex Mirror:** Focal length (f) and radius of curvature (R) are positive.

4.Refraction of Light



- **Definition**
  - Bending of light when passing from one medium to another due to a change in speed.
- **Laws of Refraction (Snell's Law)**
  - (constant for a given medium pair) and a definite colour.
    $\frac{\sin i}{\sin r} = n$
  - Incident ray, refracted ray and normal, all in the same plane.
- **Refractive Index**
  - Absolute Refractive Index,
    $n = \frac{\text{Speed of light in vacuum } (c)}{\text{Speed of light in medium } (v)}$

5. Refraction by Lenses



- **Lens**
  - A transparent material bound by curved surface that refracts light.
- **Types of Lenses**
  - Convex Lens (Converging Lens)
    - Thicker at the centre; converges light to a focal point.
  - Concave Lens (Diverging Lens)
    - Thinner at the centre; diverges light as if originating from a focal point.
- **Image Formation by Lenses**
  - Convex Lens
    - Forms real/inverted or virtual/erect images depending on object position. Real image may be erect and magnified or diminished. Virtual image always magnified.
  - Concave Lens
    - Always forms virtual, diminished images.
- **Lens Formula and Magnification**
  - Lens Formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$
  - Magnification Formula $m = \frac{h'}{h} = \frac{v}{u}$
  - Magnification is positive for virtual images ,and negative for real images.

6. Power of a Lens
- **Definition**
  - The ability of a lens to converge or diverge light.
- Formula
  $P = \frac{1}{f \text{ (in metres)}}$
- Power of Combined Lenses
  $P_{\text{total}} = P_1 + P_2 + P_3 + \ldots$